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1 Appendix 1: Alternative partial function semantics [unrelated to the rest of the article]

1.1 Example of a proof (FARMER'S CALCULUS)

Partially based on WILLIAM M. FARMER & JOSHUA D. GUTTMAN's article "A Set Theory with Support for Partial Functions".
For simplicity of understanding by the reader, the notation from the article is used here.

Definition 1.1. Value of a function.

$$f(x) \simeq (\iota y : \langle x, y\rangle \in f)$$

Theorem 1.1. If f is univocal, then

$$(f(x) \simeq y) \longleftrightarrow (\langle x, y\rangle \in f)$$

Proof.
($\Leftarrow$) (Obvious part) Here we use axiom PFOL10′ which is equaivalent to PFOL10:

$$(\iota x : \varphi)\downarrow \quad \overset{\text{PFOL10}'}{\Longleftrightarrow} \quad (\exists! x : \varphi)$$

So the proof is the following:

$$(\langle x, y\rangle \in f) \quad \overset{\exists\text{-intro}}{\Longrightarrow} \quad (\exists w : \langle x, w\rangle \in f)$$

$$f \text{ univocal} \quad \overset{\text{def}}{\Longleftrightarrow} \quad (!w : \langle x, w\rangle \in f)$$

$$(\exists w : \langle x, w\rangle \in f) \quad \& \quad (!w : \langle x, w\rangle \in f) \quad \overset{\exists!\text{-intro}}{\Longrightarrow} \quad (\exists! w : \langle x, w\rangle \in f)$$

$$(\exists! w : \langle x, w\rangle \in f) \quad \overset{\text{PFOL10}'}{\Longrightarrow} \quad (\iota w : \langle x, w\rangle \in f)\downarrow$$

$$(\iota w : \langle x, w\rangle \in f)\downarrow \quad \overset{\text{PFOL11}}{\Longrightarrow} \quad \langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f \quad \& \quad (\langle x, y\rangle \in f) \quad \overset{f \text{ univocal}}{\Longrightarrow} \quad (\iota w : \langle x, w\rangle \in f) \simeq y$$

$$f(x) \simeq y$$

($\Rightarrow$)

$$(y\downarrow) \quad \text{(by PFOL8)}$$

$$(y\downarrow)\ \ \&\ \ ((\iota w : \langle x, w\rangle \in f) \simeq y) \quad \overset{\text{PFOL7}}{\Longrightarrow} \quad (\iota w : \langle x, w\rangle \in f)\downarrow$$

$$(\iota w : \langle x, w\rangle \in f)\downarrow \quad \overset{\text{PFOL11}}{\Longrightarrow} \quad \langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f$$

$$\langle x, (\iota w : \langle x, w\rangle \in f)\rangle \in f \ \ \&\ \ ((\iota w : \langle x, w\rangle \in f) \simeq y) \quad \overset{\text{PFOL7}}{\Longrightarrow} \quad (\langle x, y\rangle \in f)$$

Done.

Theorem 1.2.

$$f(x)\downarrow \longleftrightarrow\ x \in \{x | \exists! y : \langle x, y\rangle \in f\}$$

Proof.
Done.
By extension of definitions from

$$\forall f \forall x \exists! y : (x \in \mathrm{dom}(f)) \wedge (f \text{ univocal at } x) \to (x, y) \in f$$

we have

$$\forall f \forall x (x \in \mathrm{dom}(f)) \wedge (f \text{ univocal at } x) \to (x, f(x)) \in f$$

or from: (but it backward implication can be proven without such complifications)

$$\forall f \forall x \exists! y : (f \text{ univocal at } x) \to ((x \in \mathrm{dom}(f)) \iff (x, y) \in f)$$

or maybe 'Extension by new constant and function names'?

In general, a closed formula

$$\forall \vec{x} : \exists! y : \varphi$$

implies

$$\forall \vec{x} : \varphi[(Iy : \varphi)/y]$$

. IMHO, the most important part here is that y is not free in the context below. (the reason is identical to the one in the Generalization Rule) YES! Important: get rid if quantifier over arguments. If the context doesn't have y, the substitution of "whole phi" would be already done.
something that a lil bit concerning is that context is not seen inside I, but it may not be important.
So the following rules are proposed:

$$\frac{\Gamma \vdash \exists! y : \varphi}{\Gamma \vdash \varphi[(\iota y : \varphi)/y]} \text{ ExtByDef (if } y \notin \mathrm{FV}(\Gamma))$$

and

$$\frac{\Gamma \vdash \exists y : \varphi}{\Gamma \vdash \varphi[(\epsilon y : \varphi)/y]} \text{ ExtByCons (if } y \notin \mathrm{FV}(\Gamma))$$